



Instructions

Please write your name in CAPITALS and complete the Candidate number in the spaces below.

Use a PENCIL (B or HB)

Rub off any answers you wish to change with an eraser.

No calculators, mobile phones, smartwatches, or any electronic devices are permitted.

Show all workings clearly in the answer booklet provided.

Maintain **silence** during the exam. Raise your hand if you need assistance.

Candidate Name			
Country	NIGERIA	Class	YEAR 10
CRC	EMC25001	Candidate Roll No	

NATIONAL ROUND

- 1) An object is launched from the bottom of pit 34.3 meters below the ground surface, directly upward at 39.2 m/s. The equation for the object's height h at time t seconds after launch is $h = -4.9t^2 + 39.2t - 34.3$. Where h is in meters, for how long is the object at or above the ground surface?

Factor completely. (Question 2-4)

2) $24xy^2 - 12xy + 36x^2y$.

3) $3ax^2 - 21axy + 30ay^2$.

4) $6m^2n^2 + 18m^2n - mn^2$.

- 5) A kite is 32m above the ground. The angle the kite string makes with the ground is 39° . How long is the kite string to the nearest meter?

From a class of 30 students, 12 enjoy cricket (C), 14 enjoy Netball (N) and 6 enjoyed both cricket and netball. (Question 6-8)

- 6) Illustrate this information in a venn diagram.
- 7) State the number of students who enjoy (i) netball only (ii) neither cricket nor netball.
- 8) Find the probability that a person chosen at random will enjoy: (i) netball (ii) netball only (iii) both cricket and netball.

The area of rectangle lot is $3x^2 + 8x + 4$ (Question 9-10)

9) Determine the dimension of the lot

10) If the of value x is 4meters, state the dimensions and the area of the lot.

Complete the square for each of the following; then state the vertex for each.

(Question 11-12)

11) $y = x^2 + 4x - 3$

12) $y = -3x^2 - 6x + 2$

13) Express $(3x^2 y^4) (4x y^2)$ in its simplest form.

14) Mark and Matthew's teacher has asked that they use their own rule to construct a sequence of number, starting with 5. The sequence that they have constructed are given below.

Matthew's sequence: 5; 9; 13; 17; 21; ...

Mark's sequence: 5; 125; 3125; 78125; 1953 125; ...

Write down the n th term (or the rule in terms of n) of:

- Matthew's Sequence
- Mark's sequence

The following sequence of numbers forms a quadratic sequence; -3; -2; -3; -6; -11; (Question 15-16)

15) Determine an expression for the n^{th} term of the quadratic sequence.

16) Calculate the first difference between the 35^{th} and 36^{th} terms of the quadratic Sequence.

Solve the following quadratic equations by factorizing. (Question 17-18)

17) $a^2 - 6a + 9 = 0$

18) $\frac{x^2}{2} + \frac{7x}{4} = 0$

Given $\sum_{r=0}^{99} (3t - 1)$

19) Write down the first three terms of the series.

20) Calculate the sum of the series.